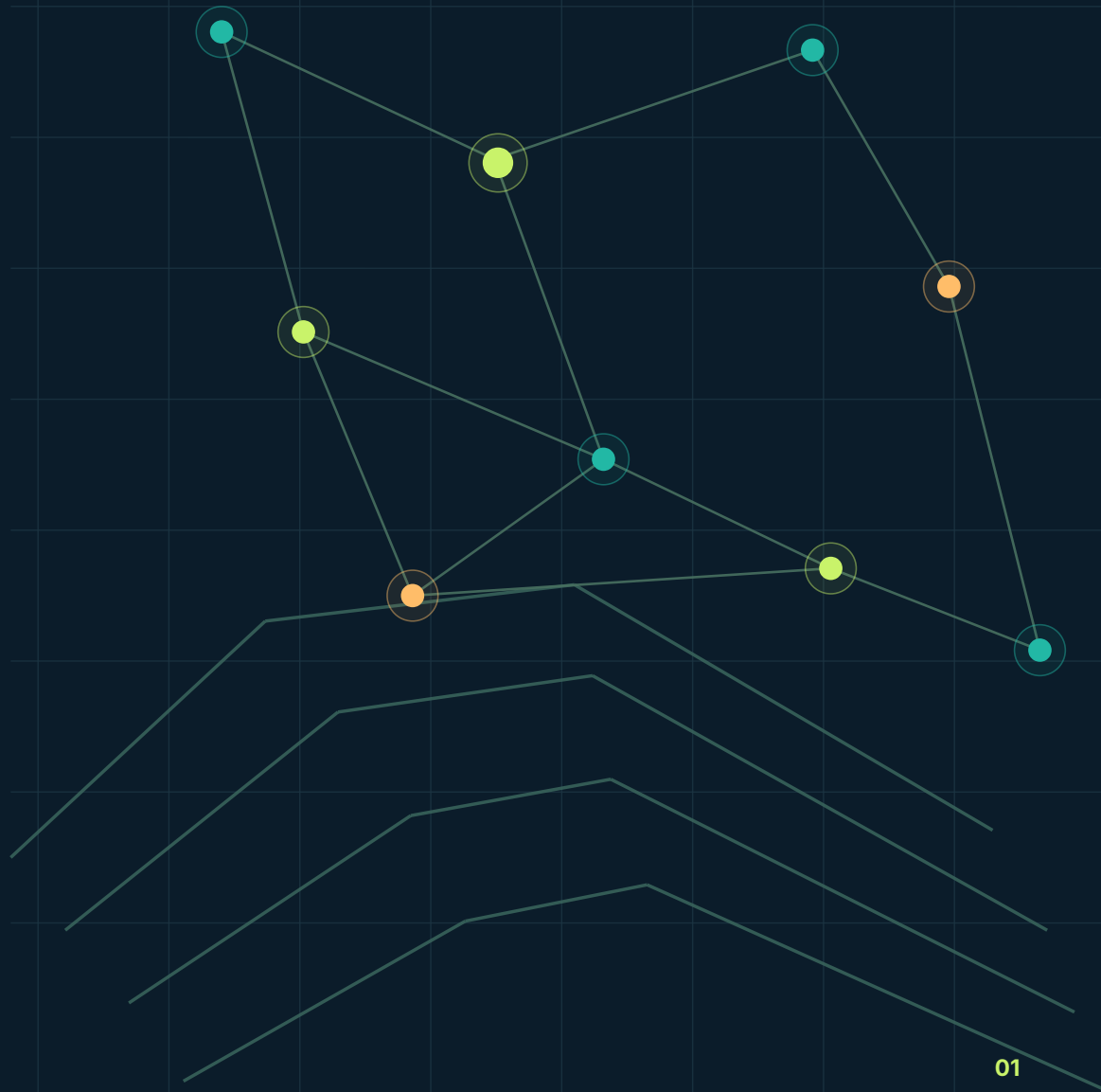
RESEARCH PROPOSAL • INDIA AGRICULTURE

Predict the harvest. Test what drives it.

A district-level framework for crop-yield prediction, causal analysis and policy insight.



THE CENTRAL LOGIC

Predict first. Identify causes carefully. Forecast prices only if the data are ready.

Causal and Predictive Modeling of Indian Crop Yields and Agricultural Prices

When harvest uncertainty travels downstream

Agricultural forecasts matter because decisions are made before the final harvest is known.

~1 in 5

of national income

Agriculture & allied activities

46.1%

of the workforce

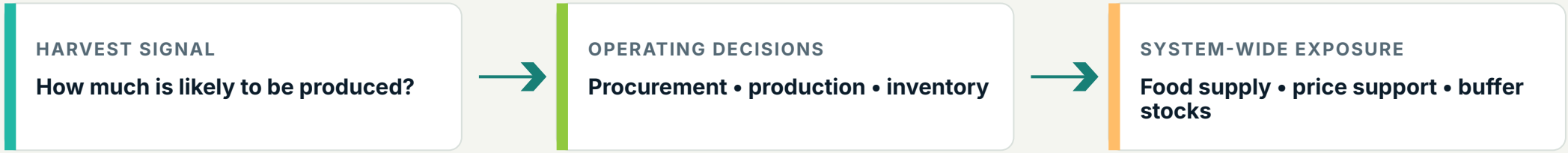
Dependent on agriculture & allied activities

$382B

food-processing market

India • FY2024–25 estimate

A forecast is useful when it changes a decision—not just when it scores well.



Sources: Government of India, Economic Survey 2025–26, Ch. 6; IBEF, Food Processing Industry (FY25).

One framework. Three distinct questions.

Keep prediction, causal explanation and price forecasting analytically separate.

What information improves district-level crop-yield forecasts—and what can the data credibly tell us about why yields change?

01

PREDICT

What yield can we forecast?

Benchmark history-only models, then test what weather, inputs and structural variables add.

02

IDENTIFY

What actually changes yields?

Estimate effects only for defined interventions with credible counterfactuals.

03

EXTEND

Can prices be forecast too?

Proceed only after crop, market and time coverage pass the data-quality audit.

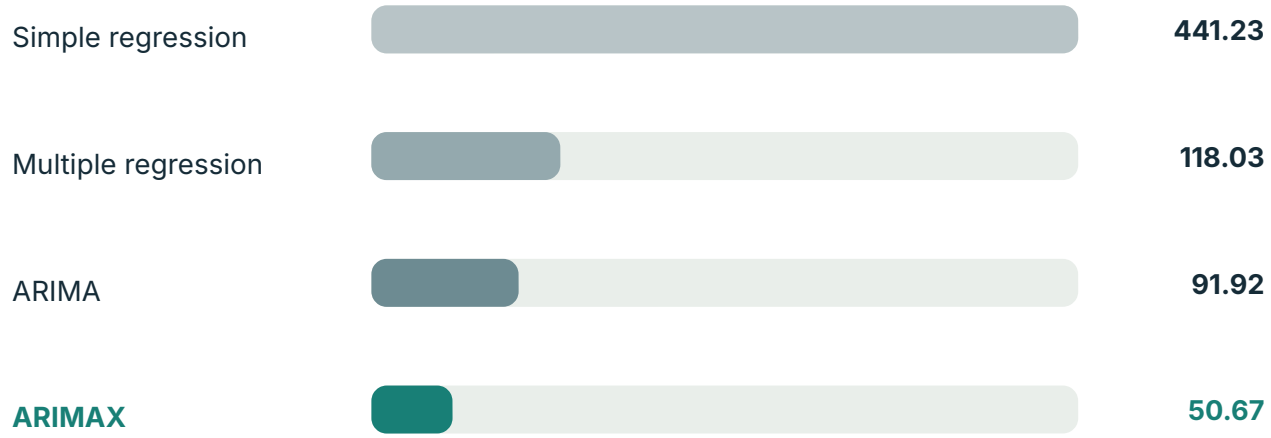
The committed deliverable is a validated yield-prediction pipeline—not a pre-promised causal effect or price forecast.

Weather-aware forecasting has a signal—but the evidence is local

Published results motivate the design; they are not performance results from this project.

EXTERNAL BENCHMARK • PEARL MILLET, ALWAR

Mean absolute error (MAE) • kg/ha • lower is better



Train: 1997–2013 | Holdout: 2014–2016 | One crop, one district

CLIMATE CONTEXT

Higher maximum temperatures were associated with yield losses.

−15%
Rice yield

−22%
Wheat yield

Conditional projections to 2100 under the study's significant climate-change scenario—not unconditional forecasts.

Sources: Dharmaraja et al. (2020), Tables 5–6; Birthal et al. (2014). The project has not yet generated its own model baseline.

The contribution is a disciplined test—not another model name

Three open questions connect the literature to a reproducible empirical plan.

GAP 01

Incremental value & portability

Do added variables beat a historical-only baseline—and across which crops and regions?

NESTED BENCHMARKS

Same future-year holdout; report crop and regional variation.

GAP 02

Prediction ≠ causality

Feature importance or correlation cannot prove irrigation, roads, credit or policy caused yield changes.

CREDIBLE COUNTERFACTUALS

Define one intervention and test design assumptions.

GAP 03

Policy & price data feasibility

Market prices and policy text are not automatically aligned with district-year outcomes.

AUDIT BEFORE EXTENSION

Validate coverage; structure and review policy attributes.

Expected contribution: an auditable answer to what helps, where gains generalize, and what the data can actually identify.

Methodological framing: Jabed & Murad (2024); Angrist & Pischke (2009); Imbens & Rubin (2015).

A broad district-level dataset—with comparability checks still open

Working inventory spans crop outcomes, biophysical conditions, economic structure and policy text.

26	~400K	18,060	34 years	602
source CSV files	cumulative source rows	rows in anchor crop file	maximum coverage	district entries • 20 states

FOUR INPUT LAYERS

01 CROP OUTCOMES

Yield • production • area

02 BIOPHYSICAL

Rainfall • temperature • soil

03 AGRO-ECONOMIC

Inputs • irrigation • wages • roads • livestock

04 POLICY TEXT

Central and state documents

IMPORTANT RECONCILIATION Project inventory: 602 district entries. Current ICRISAT/TCI pages describe 571 districts, 27 datasets and 1,030 variables. These counts may reflect different source vintages or geographic definitions; reconcile before defining the final panel.

Sources: project working-file inventory; ICRISAT District Level Data; Tata-Cornell Institute data page. Counts are not yet final unique panel observations.

Integrated is not the same as inference-ready

The first-pass pipeline is in place; the next gate is a reproducible quality audit.

COMPLETED OPERATIONS

- Standardized source tables using Python / Pandas
- Integrated against the anchor crop file; considered join cardinality
- Recoded source sentinel −1 to missing
- Applied variable-level mean / median imputation
- Prepared 175-column yield + weather and 174-column yield + structural matrices

MUST BE VALIDATED BEFORE MODELING

01 KEYS & GEOGRAPHY
Unique crop–district–year keys; boundary and merge logs

02 MISSINGNESS
Residual gaps, outliers, alternate-imputation sensitivity

03 NO DATA LEAKAGE
Fit imputation / scaling within training years only

04 FORECAST TIMING
Use only predictors available at the forecast cut-off

05 EXTENSION LAYERS
Human-check policy extraction; align market prices

STATUS • INTEGRATION + FIRST-PASS PREPROCESSING COMPLETE; VALIDATION REMAINS OPEN

Missing-value treatment and merge operations are reported project operations; their adequacy is not yet established.

Prove the value of each feature group

Nested comparisons show what each data layer adds beyond a transparent historical baseline.

00 HISTORICAL ONLY

Last yield / trend
Simple regression or time series

01 + WEATHER & SOIL

Rainfall • temperature
Soil / growing conditions

02 + FARM INPUTS

Irrigation • fertilizer
Other agricultural inputs

03 + STRUCTURAL FEATURES

Socioeconomics • infrastructure
Livestock / market access

MODEL LADDER

Regression / ARIMA / ARIMAX → Random Forest / XGBoost → RNN, LSTM or Transformer only if the series support them.

EVALUATION RULES

Later years held out • Same test observations • MAE primary • RMSE secondary • R^2 complementary

Fit imputation, scaling and feature selection on training years only; define what is genuinely known at forecast time.

H1 **Weather / soil / inputs reduce out-of-time MAE vs. history alone.**

H2 **Structural features add forecast value beyond H1's feature set.**

Evaluation follows the proposal's time-based, common-holdout design; no model winner is assumed in advance.

Causal claims need a credible counterfactual

The causal track is separate from forecasting—and conditional on the design, not just the dataset.

District + year fixed effects control for stable district traits and common year shocks; they do not, by themselves, establish causality.

POLICY TEXT → STRUCTURED FEATURES

Policy documents — **LLM-assisted extraction** — **Human review** — **District-year features**

Extract: scheme • instrument • dates • eligibility • geography • implementation intensity

DESIGN GATE

Difference-in-Differences or regression discontinuity only when timing, comparison groups / thresholds and identifying assumptions are defensible.

H3 • EXPLORATORY

Market connectivity / credit access

Test whether a plausibly exogenous improvement raises yield; road density and bank counts alone are imperfect proxies.

H4 • POLICY EFFECT

A defined intervention and yield outcomes

Primary: mean yield. Secondary: yield variability. No assumption that all policies or post-2017 changes share one effect.

Method references: Angrist & Pischke (2009); Imbens & Rubin (2015). Policy-document presence is not, by itself, a measured treatment.

A staged route to decision-grade evidence

Each step has a clear gate; the project only advances when the evidence is ready.

A

AUDIT

Resolve keys, geography, coverage, units, missingness and imputation sensitivity.

Deliverable: quality report

B

PREDICT

Run time-based benchmarks; quantify gains from each feature group across crops and regions.

Deliverable: benchmark matrix

C

IDENTIFY

Choose one policy / exposure and test whether a credible counterfactual exists.

Deliverable: design + diagnostics

D

EXTEND

Forecast prices only if crop-market-time coverage and forecast-time availability pass audit.

Deliverable: go / no-go decision

The goal is not to declare a model winner before testing.

It is to show what improves forecasts, where the gains generalize, and which causal claims the data can support.

Selected references

Core sources behind the motivation, external benchmark, data foundation and identification strategy.

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Jabed, M. A., & Murad, M. A. A. (2024).

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Angrist, J. D., & Pischke, J.-S. (2009).

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Imbens, G. W., & Rubin, D. B. (2015).

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All reported model MAEs are external literature results; the assembled project dataset has not yet produced its own benchmark scores.